NEED OF SIMULATION

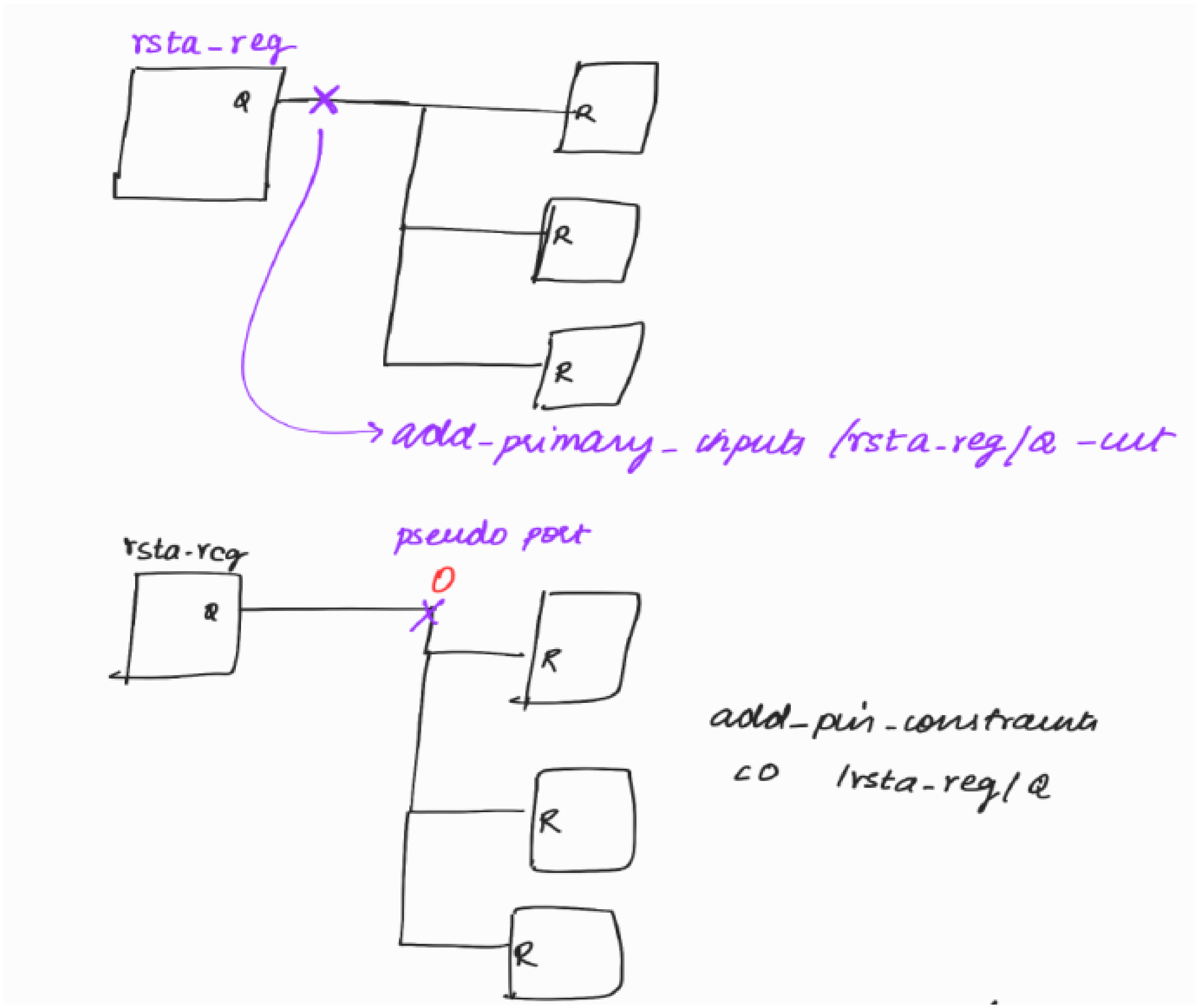
To overcome the limitations of ATPG tool

- Internal cut-points

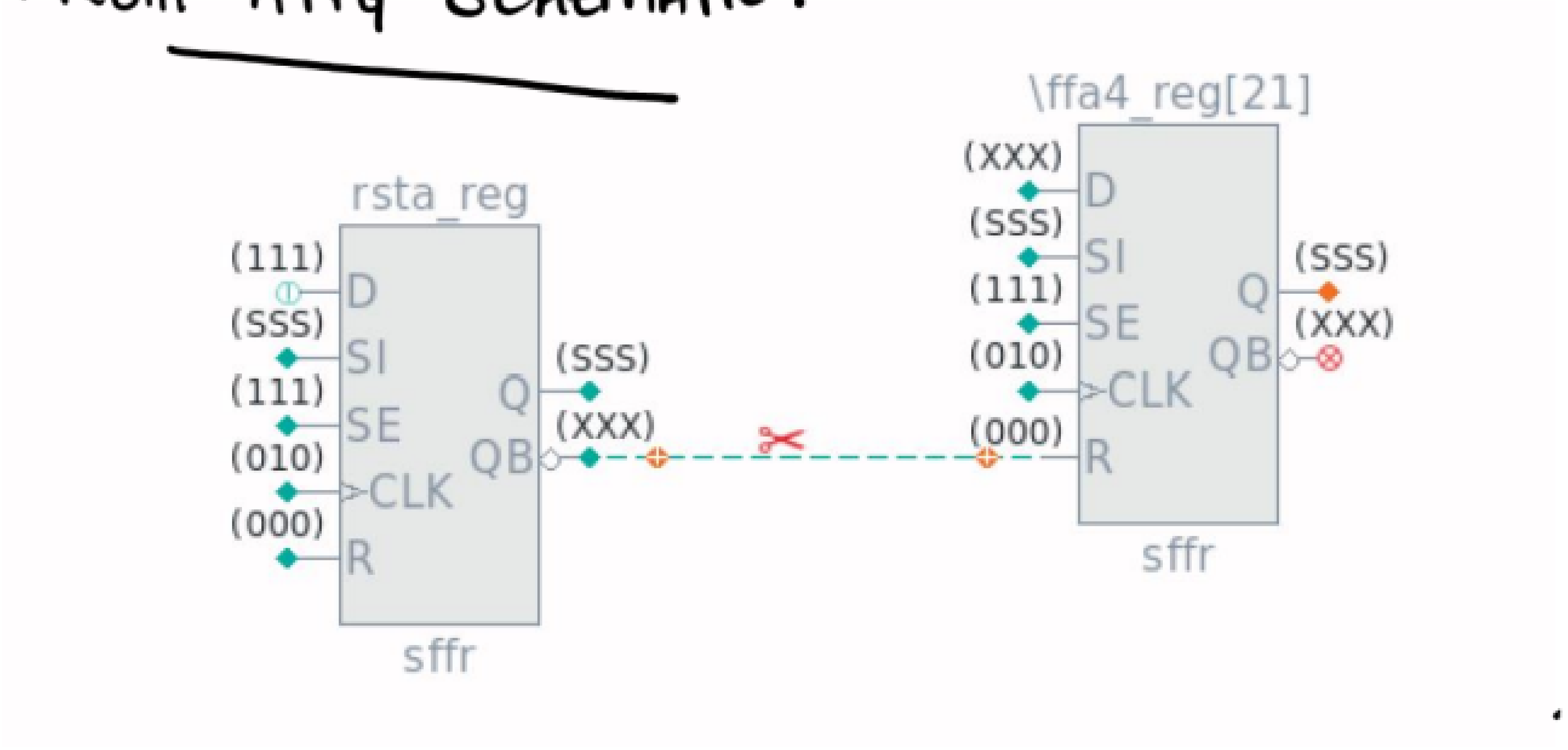
They are used to clear off DRCs temporarily.

But before the patterns is given to the tester, all these cut-points have to be removed. Because tester can apply patterns on EDT channels. It can't apply a value on an internal pseudo port.

ATPG tool will write a pattern by considering the internal cut-point.
If the same patterns is given to the tester, our patterns will fail on the tester. Since it can't apply a value on an internal pseudo port.
So in order to detect these kind of issues, we need to do simulation.



FROM ATPG SCHEMATIC:

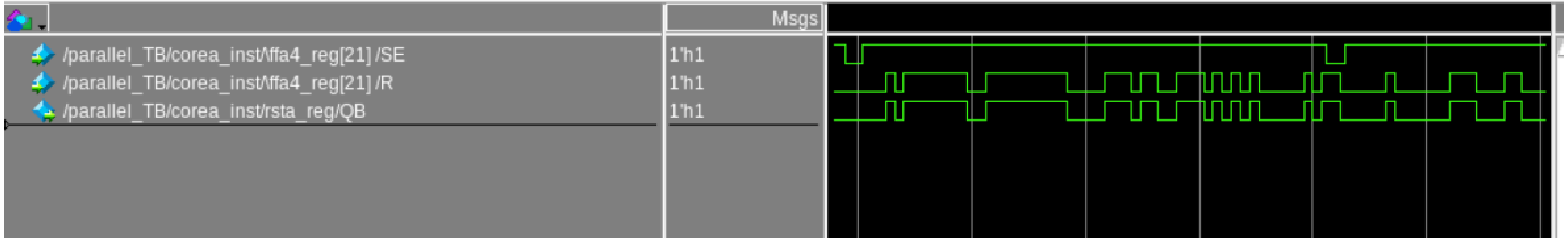


ATPG tool has written a pattern by considering the cutpoint and by considering the value at the cutpoint.

∴ In ATPG, rst of the ffs are getting 0.

If the same patterns is given to the tester, our patterns will fail on the tester. Since it can't apply a value on an internal pseudo port.

So we need to do simulation. If we have used internal cut-points, simulations will also fail. Because simulation tool also can't apply a value on an internal pseudo port. At the time of simulation, rsta_reg/QB is controlling the reset of the flop. So reset of the flops are toggling. So it will fail.



QUESTA SIM WAVEFORM

NOTE :
Internal cut-points can be used only during temporary releases.

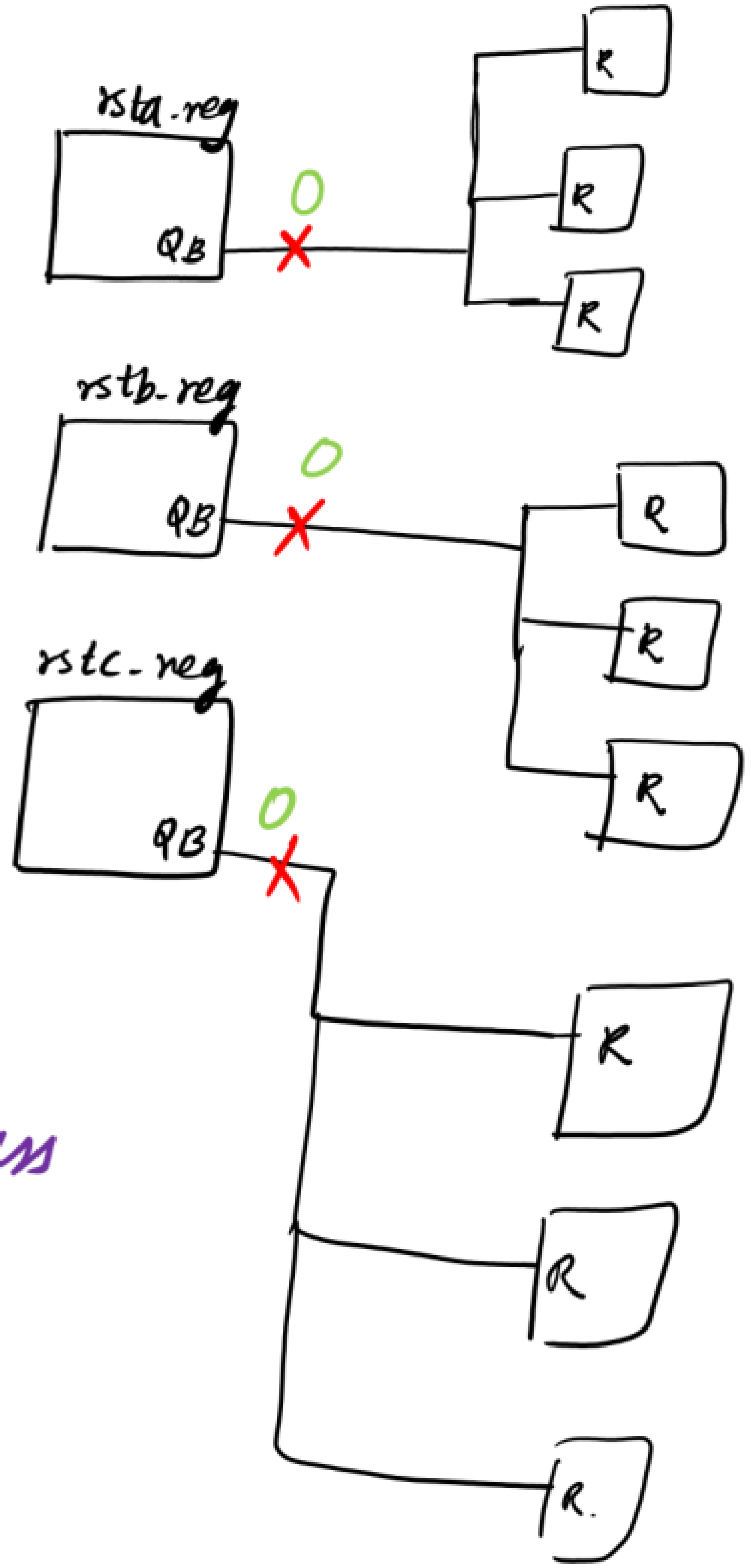
But before the patterns is given to the tester (during final release), all these cut-points have to be removed. Because tester can apply patterns on EDT channels. It can’t apply a value on an internal pseudo port. So patterns will fail on the tester.
In-order to ensure that we have not used any internal cut-points during final release, we can run simulations.

During temporary releases, we can false pass the simulation.

Internal cutpoints:

```
add_primary_inputs /rsta_reg/QB -cut
add_primary_inputs /rstb_reg/QB -cut
add_primary_inputs /rstc_reg/QB -cut

add_pin_constraints C0 /rsta_reg/QB
add_pin_constraints C0 /rstb_reg/QB
add_pin_constraints C0 /rstc_reg/QB
```



At the place of the cutpoint, a pseudo port will be formed.

At the pseudo port, we can force the desired value.

Used the internal cutpoint.

Our patterns will fail during simulation.

If it is temporary release, we can false pass the simulation

For temporary releases, we can false pass the simulation by forcing the required value. And simulation tool doesn't understand about pseudo port. So it will throw an error that 'pport' is not found. So we need to remove all the pseudo ports from the testbench.

(i) Generated patterns with internal cutpoints.

(ii) Run the simulation

Error :- pport (pseudo port) is not found

Reason :- QuestaSim tool doesn't understand about pseudo port.

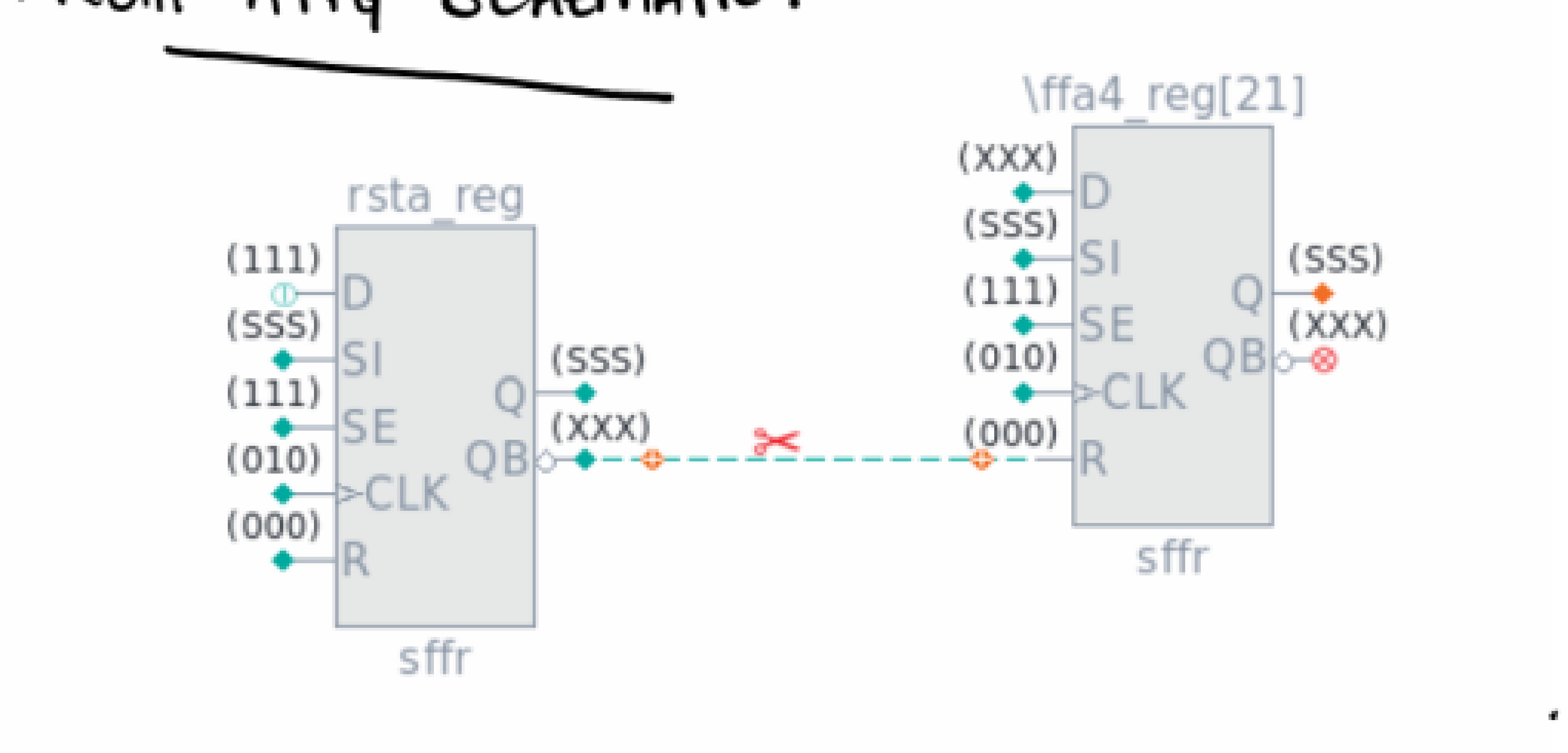
Soln :- Remove the pseudo ports from the testbench.

(iii) Again run the simulations

Simulations is failing (Mismatches)

Reason for simulation mismatches:

FROM ATPG SCHEMATIC:



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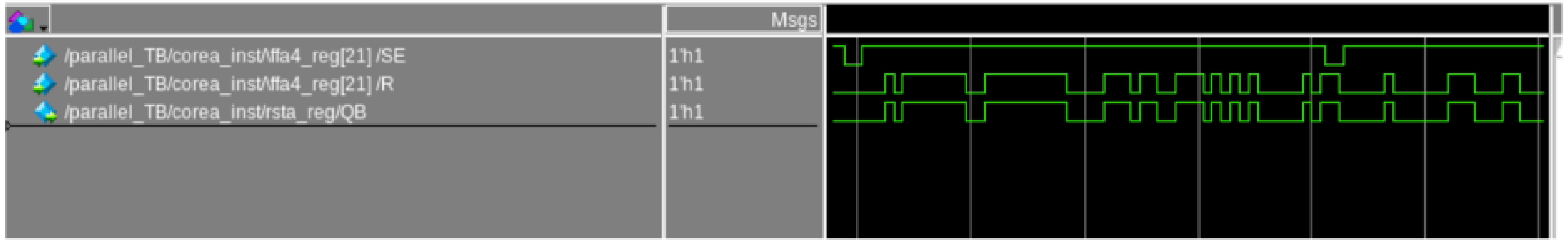

QUESTA SIM WAVEFORM

Similarly for rstb.reg and rstc.reg also.

Soln :-

For temporary releases, we can false pass the sim by forcing the required value.

i.e; using force command, we can force 0 at QB of rsta_reg, rstb_reg, rstc_reg.

```
vsim -c  -debugDB -voptargs=+acc parallel_TB -do runsim.do -c +nospecify -l simulation.log_class
```

```
add wave -r /*;
force -freeze parallel_TB/corea_inst/rsta_reg/QB 0;
force -freeze parallel_TB/corea_inst/rstb_reg/QB 0;
force -freeze parallel_TB/corea_inst/rstc_reg/QB 0;
run -all;
```